SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation   COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Total Marks:20   Duration : 1 Hour

Scenario Based

Code of Conduct:

*I **Shaik Mohammed Waseem Rayan (192373004)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

Shaik Mohammed Waseem Rayan

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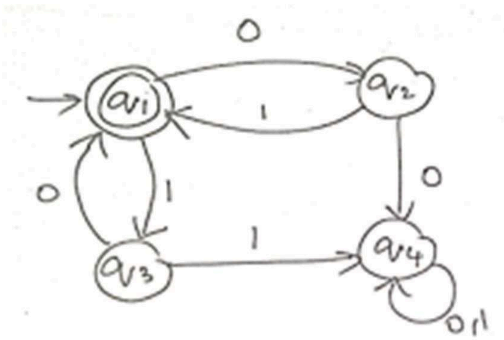
1. Given a DFA used for validating a specific PIN pattern bbbc, bca ,c, ca, caaaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

2. Demonstrate how a given finite automaton can be converted into a regular expression



3. Consider the following

grammar S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

4. I)Show that if L1 and L2 are regular then (L1+l2) also regular Ii)Is the language { a^p| p is prime} regular? Justify

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1) Minimize the DFA for PIN patterns PINs: bbbc, bca, c, ca, caaaaaa **Minimized DFA (States)**

State	**Meaning**	**Input a**	**Input b**	**Input c**	**Final**
q0	Start	q4	q1	q3	No
q1	Seen b	Dead	q2	q5	No
q2	Seen bb	Dead	q6	Dead	No
q3	Seen c	q4	Dead	Dead	**Yes** (c)
q4	Seen ca…	q4	Dead	Dead	**Yes** (ca, caaaaaa)
q5	Seen bc	q4	Dead	Dead	**Yes** (bca)
q6	Seen bbb	Dead	Dead	q7	No
q7	Seen bbbc	Dead	Dead	Dead	**Yes** (bbbc)
Dead	Trap	Dead	Dead	Dead	No

State Reduction Benefits

- Removes equivalent/redundant states.
- Reduces memory required to store transition tables.
- Faster state transitions, improving execution speed.
- Saves power and processor time in embedded systems.
- Simplifies implementation and debugging.

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2) Convert the given FA to Regular Expression

From the diagram:

- Start state: q1
- Final state: q1
- q1 ↔ q2 on **0,1**
- q1 ↔ q3 on **0,1**
- q2 → q4 on **0**
- q3 → q4 on **1**
- q4 has loop on **0,1**

The accepted strings are those that start and end at q1.

Regular Expression

$$((0 + 1)(0 + 1))^*$$

or equivalently,

$$(00 + 01 + 10 + 11)^*$$

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3) Grammar
Given
S → aSc | T | U | ε
T → bTc | ε
U → aUb | ε
(A) Language Defined
The grammar generates

- Equal number of **a's** and **c's**
- Equal number of **b's** and **c's**
- Equal number of **a's** and **b's**

Hence,

$$L = \{a^n c^n\} \cup \{b^n c^n\} \cup \{a^n b^n\}$$

including ε.

(B) Derivation Tree for aaabbcc

```
S
|
a
|
S
|
a
|
S
|
a
|
U
├── b
|
U
├── b
|
ε
```

(Equivalent leftmost derivation)
S
→ aSc
→ aaScc
→ aaaUcc

→ aaaUbcc
→ aaabbcc

(C) Is the grammar ambiguous? Yes.

A grammar is ambiguous if one string has two different parse trees. Since **ε** can be generated from

S→ε

S→T→ε

S→U→ε

there are multiple derivations for ε.

Hence, **the grammar is ambiguous.**

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4) Show that if L_1 and L_2 are regular then ($L_1 + L_2$) is regular Proof

Let

- DFA M_1 recognize L_1
- DFA M_2 recognize L_2

Construct an NFA with a new start state having ε-transitions to the start states of M_1 and M_2**.** This NFA accepts

$$L1 \cup L2$$

Since every NFA has an equivalent DFA, the union language is also regular. Therefore,

$$L1 + L2$$

is **regular.**

4(ii) Is $L = \{a^p \mid p$ is prime$\}$regular? Answer No.

Justification (Pumping Lemma) Assume the language is regular. Let pumping length be **m.** Choose

$$w = a^q$$

where **q** is a prime number greater than m. By the pumping lemma,

$$w = xyz$$

with

- $|xy| \le m$
- $|y| > 0$

Pump y twice:

$$xy^2z$$

Its length becomes

$$q + |y|$$

or by choosing an appropriate pumping multiple,

$$|y|$$

$$q)$$

$$q + (k-1)\,|y$$

$$| = q\,(1 +$$

which can be made **composite**, so the pumped string is **not** in the language. This contradicts the pumping lemma.

Hence,

$$\boxed{L = \{a^p \mid p \text{ is prime}\}}$$

is **not regular.**